Speech to Braille Language encoding for Blind people

A PROJECT REPORT

Submitted by

FAIZAN B PATEL (16EL010)

KRINA B SHETH(17EL0315)

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Birla Vishvakarma Mahavidyalaya Engineering College (V.V.Nagar)

(An Autonomous Institute)

Electronics Engineering

2019-2020

CERTIFICATE

Date:-10/07/2020

This is to certify that the dissertation entitled "**Speech to Braille Language encoding for Blind people**" has been carried out by **Faizan Patel and Krina Sheth** under my guidance in fulfillment of the degree of Bachelor of Technology in Electronics Engineering (8^{th}Semester) of Gujarat Technological University, Ahmedabad during the academic year 2019-2020.

Guides: - Prof. AjaykumarDaiya (Faculty Guide)

Mr. MayankChavda (Industrial Guide)

Head of the Department

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ABSTRACT

This project presents the design and implementation of an electronic device that can convert the input voice from mobile application to Braille code and display it on LCD. This electronic device can be a media for learning to children with visual impairment or blind people can able to write it independently without help of anything. The main aim of this Project is to reduce the gap between blind and sighted people. I.e. due to this method the blind people can easily learn and reading, writing.

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1. INTRODUCTION

1.1 Introduction

The term written info is employed here to incorporate all print material (mainly on paper however conjointly on labels, signs or tags), electronic material (stored in info systems or WWW content and displayed in monitors or projected on screens) and written material that embrace text, scientific symbols and graphical representations (e.g. diagrams, maps, pictures, drawings). Once the written info is language-based (i.e. the text), it's easy to state that speech (i.e. the oral kind of language), is that the direct different or substitute modality to convey the data to a visually handicapped person and it's terribly useful for the part hawk-eyed persons.

Braille is a vital language employed by the blind to browse and write. It's important for communication and academic functions. Braille shows many blessings for blind folks over alternative writing systems. First, it's versatile, creating it attainable to represent alphabet letters, accentuated letters, punctuation, numbers, and even musical notes. Second, the reduced range of keys needed to input a chord permits an on the spot finger-key match, therefore eliminating visual/tactile scanning. Therefore, blind users are able to do high text-entry rates employing an easy seven key chord-keyboard. However, Braille acquisition has been declining principally because of the employment of text and helpful software package, like screen readers. Moreover, with personal computers, blind users were nudged to use ancient QWERTY keyboards, since hardware diversifications were too dearly-won and cumbersome. Four of these reasons, alongside the large effort needed to effectively browse and write Braille, it's been increasingly neglected. However, Braille acquisition continues to be the foremost empowering kind of acquisition for blind folks and has shown to be the strongest predictor of upper levels of education and employment in adult life.

1.2 Aim of the project

This project presents the look associated implementation of a device which will convert the input voice from mobile application to Braille code and show it on liquid crystal display.This device are often a media for learning to youngsters with disablement or blind folks will able to write it severally while not facilitate of something.

- Blind folks will browse, speak and listen to simply however they can't write.
- Therefore this device can facilitate them to jot down terribly simply by spoken communication the words in mobile.

1.3 Problem Specifications

Some of the foremost limitations that persons with visual impairments face are:

- To access written information,
- To control devices with complicated user interfaces.
- To induce orientation and quality support
- To look at TV broadcastings, movies, live performances or shows.

This report shows the matter of writing the Braille in speed and printing the dot language simply for the blind.

Current panorama. Although participants learned Braille in numerous establishments, the method was similar. Once associate introduction to the Braille cell and dots positions with well-known artifacts (e.g. a 6-egg box), the training sequence typically starts with reading. Yet, associate overlap between reading and writing might occur.

Braille usage, even for mundane functions, is far larger for specialists than for novices. The two instructors were a lot of awake to Braille labeling in several contexts, like market product and elevator buttons, contrastive with novice participants that over-relied on family facilitate or reminiscences before visual defect. The foremost good participants additionally resorted to Braille in things requiring privacy, like making notes as teaching aids and labeling their artifacts for future use (ID/bank cards).

Lack of interactivity. All participants took or presently teach categories in tiny teams (2-4); however the interactivity among students is incredibly restricted. In advanced categories, students might act to write/read completely different components of the text. Solely sometimes, these 2 tasks could also be interactive (e.g. one writes what the opposite reads).Lack of resources. Braille categories typically confine blind learners to a Perkins Braille character printer (or similar; or maybe a slate) and to paper sheets. What's a restricted setting turns into the absence of resources once they leave the category.Current tools that permit the follow and production of Braille content square measure terribly pricey. Moreover, Braille resources (e.g. books) square measure promptly rejected, as affordable performances need higher proficiency levels.

Lack of perceived purpose. Late Braille learners usually have lower proficiency levels and don't perceive the aim of up it. Above all, they feel that audio-based technologies satisfy to produce matter feedback which the massive effort needed being good in Braille isn't advantageous. In distinction, the good ones were terribly positive regarding the benefits that such data provides in their daily lives. Lack of motivators.

Lack of perceived purpose results in low motivation levels. This contributes insufficient learning effectiveness and to the abandonment of Braille once category completion. Each instructors acknowledged the dearth of use of external motivators; however, one in every of them introduced a straightforward game with cards and proverbs that perceived to improve for the most part the students' motivation. Even those reluctant to be told Braille, expressed excitement to participate and while they were unaware of the actual fact, they were active and enhancing their Braille reading skills.

Access to data is that the 1st hurdle. Once accessible materials square measure created offered, management of data will take longer for a student with visual disabilities. It will take longer to use various access techniques to, as an example, do analysis on-line, and write a paper so produce the correct final format.Many students in academically difficult programs realize themselves finding out long hours to stay up.

1.4 Literature Review

What is Braille?

Braille may be a tactile reading and orthography utilized by blind and visually impaired folks that cannot access print materials. It uses raised dots to represent the letters of the print

alphabet. It additionally includes symbols to represent punctuation, arithmetic and scientific characters, music, pc notation and foreign languages.

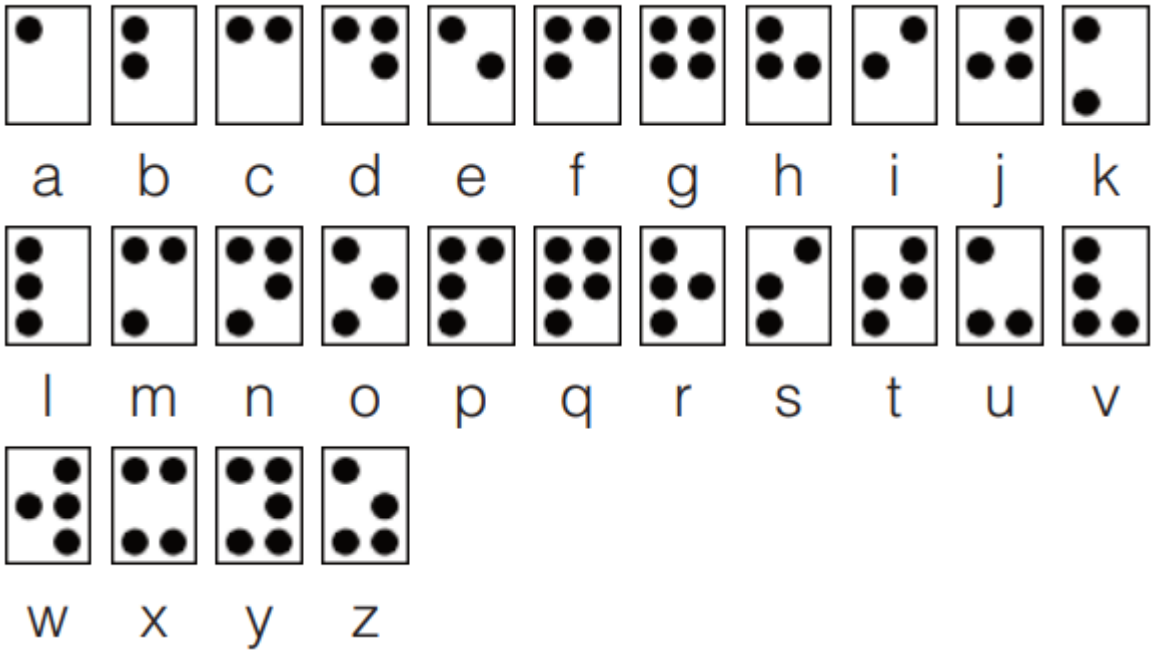

Fig 1.1

Why is Braille used?

Braille isn't a language.It is a code by that all languages is also written and skim.Through the employment of Braille, those who are blind are able to review and study the word.It provides a vehicle for acquisition and provides a personal the power to become acquainted with orthography, punctuation, paragraphing and different data formatting issues.

Who uses Braille?

Braille is employed principally by those who are blind or deaf blind.It is critically vital to the lives of visually impaired individuals because the ability to browse and write in Braille opens the door to acquisition, intellectual freedom, civil right, and private security.Teachers, oldsters et al. United Nations agency aren't visually impaired tend to browse Braille by sight instead of by bit.

What does Braille look like?

Braille symbols are fashioned among units of area referred to as Braille cells.A full Braille cell consists of six raised dots organized in 2 parallel vertical columns of 3 dots (like the quantity half dozen on a dice).The dot positions are known by numbers one through to 6.63 combos are attainable exploitation one or additional of those six dots.Cells are often wont to

represent a letter of the alphabet, number, and punctuation, a part of a word or perhaps an entire word.

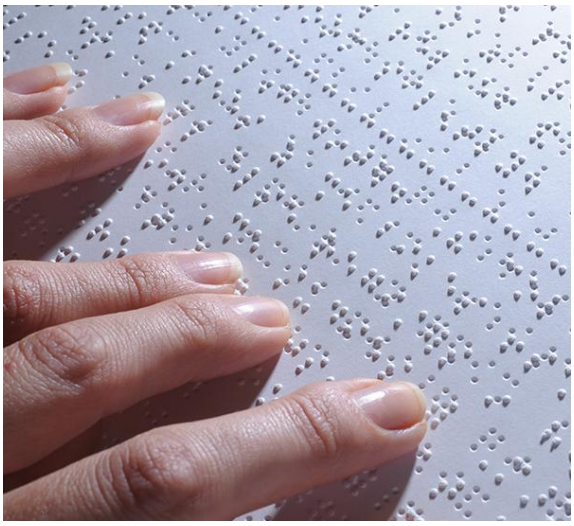

Fig 1.2

How was Braille invented?

A wide type of ways were tried to modify blind individuals to browse and write severally.Most were ways exploitation raised print letters.The prevailing belief for why the Braille system fancied by pedagogue was winning, once different ways failing, was as a result of Braille was supported a relative technique of dots, specifically designed to be known with the tip, instead of being supported symbols devised for visual recognition.The other advantage was that Braille may well be written by blind individuals fully severally.

The Braille code has undergone continuous modification over the years, notably through the addition of contractions for words that seem oftentimes in English. The use of contractions permits for quicker Braille reading and helps to scale back the scale of Braille books. Since its development in France by pedagogue within the early nineteenth Century, Braille has become an efficient means that of communication and a well-tried avenue for achieving and enhancing acquisition for those who are blind or visually impaired.

How is Braille written?

When each letter of each word is expressed in Braille, it's remarked as unconstructed Braille.Many freshly blind adults notice unconstructed Braille helpful for labeling personal or room things.Books or different reading materials may also be transcribed in unconstructed Braille.

The system usually used for reproducing textbooks and publications in English is understood as contractile Braille.In this system, cells are used separately or together with others to make a spread of contractions or whole words.For example, in unconstructed Braille the phrase "you like him" needs twelve cell areas.If it were written in contractile Braille, this same phrase would use solely six cell areas.The letters Y and L are used for the total words "you" and "like" severally.Similarly, the word "him" is created by combining the letters h and m.

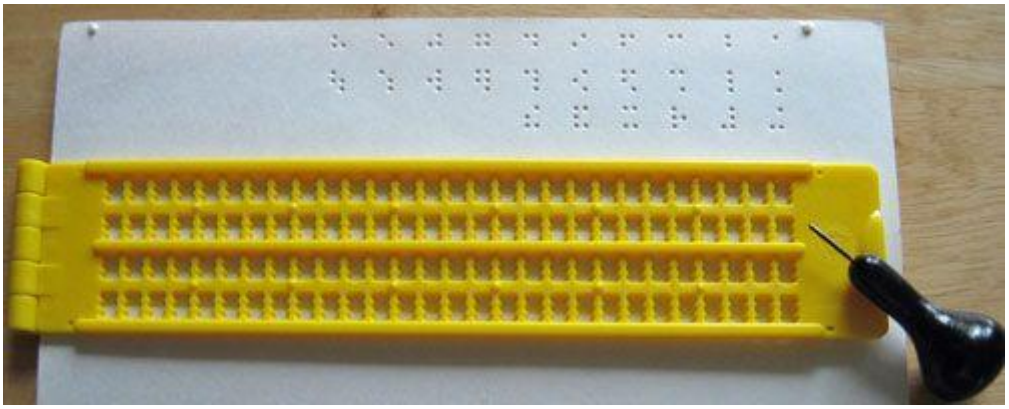

Fig 1.3

There are 180 completely different letter contractions and seventy five short kind words utilized in English contractile Braille.These short cuts cut back the quantity of paper required for reproducing books in Braille and create reading quicker.

Just as written materials are often made with paper, pencil, character printer or printer, Braille may also be written in many ways in which.Embossed Braille is typically made employing a Perkins Brailler not like a character printer that has quite fifty keys; the Perkins Brailler has solely six keys and an area bar. These keys are numbered to correspond with the six dots of a Braille cell. Since most Braille cells contain quite one dot, all or any of the brailler keys are often pushed at identical time.

Computers give and still expand extra avenues of acquisition for Braille users. Computer code programs and moveable electronic Braille note takers enable users to avoid wasting and edit their writing, have it displayed back to them either audibly or haptically and turn out a tough copy via a desktop computer-driven Braille embosser.

Writing Braille by hand is accomplished by means that of a tool known as a slate that consists of 2 metal plates hinged along to allow a sheet of paper to be inserted between them.Some slates have a wood base or guide board onto that the paper is clamped.The higher of the 2 metal plates, the guide plate, has cell-sized windows; beneath every of those, within the lower plate, are six slight pits within the Braille dot pattern.A stylus is employed to press the paper against the pits to make the raised dots.A person exploitation Braille writes from right to left; once the sheet is turned over, the dots face upward and are browse from left to right.

Braille is additionally made by special machines with six keys, one for every dot within the Braille cell.The first Braille writing machine, the Hall Braille author, was fancied in 1892 by Frank H. Hall, superintendent of the Illinois faculty for the Blind.A changed style of this device continues to be in use nowadays, as are later, similar devices. One innovation for manufacturing Braille is an electrical embossing machine just like an electrical character printer, and process system machine} processing is currently routine.

1.5 Material Required

1. PCB
2. ATMega16 Micro-controller
3. Power supply
4. Transformer

5. LCD
6. Bluetooth HC-05
7. Mains cord
8. Relay
9. Load

Software used

1. Bascom AVR Software
2. Robokits AVR USB Programmer
3. Android app: Bluetooth terminator emulator

2.Design

2.1 Analysis on vision equipments

Critical factors for the event of Braille acquisition skills

Previous analysis indicated that the age at that Braille is learned could have an effect on the speed at that Braille is processed (Mousty and Bertelson, 1985); that's, youngsters United Nations agency were schooled Braille before the age of ten usually became quicker readers.

The lack of tactual acuity could be a downside long-faced by students learning to browse through Braille and contributes to slow reading times compared to different print readers. However, through expertise and coaching, most visually impaired students will overcome this downside.

Instructional ways are known which will increase the reading rates of scholars with visual impairments. From the accessible literature, it's promptly understood that the daily instruction starting from one to 2 hours is important for the event of Braille acquisition skills. Also, the analysis has shown that visually impaired students are in danger of advancing at school with poor, fragmented Braille skills if they need not had high-quality instruction throughout their grammar school years.

Electronic Note takers

Electronic notetakers square measure little, transportable devices for storing info with the utilization of Braille or character-at-a-time printer keyboards.The keep info is also accessed through anintrinsically speech synthesizer, a Braille show, or both.In the past, similar devices that were less powerful and fewer versatile were stated as Braille note takers, that were used for storing names and phone numbers, keeping track of appointments, and taking notes.In addition to those functions, the newest electronic note takers offer advanced data processing, internet browsing, and alternative functions.

Electronic note takers permit you to urge obviate heaps of scraps of paper or container tapes you've got used for attempting to stay track of data. No electronic note taker with a screen has been developed for individuals with low vision; therefore an individual with low vision ought to take into account Associate in nursing electronic note taker with speech output. Somebody WHO is aware of Braille can most likely be better-off employing a device with Braille output.For your convenience, electronic note takers with Braille and speech output square measure listed severally here from electronic note takers with speech output solely.

This is an electronic braille notetaker.

Fig 2.1

Perkins Brailler

The Perkins Brailler could be a "Braille typewriter" with a key cherish every of the six dots of the Braille code, an area key, a backspace, and a line area key. Sort of a manual character-at-a-time printer, its 2 aspect knobs to advance paper through the machine and a printing operation lever higher than the keys.The rollers that hold and advance the paper have grooves designed to avoid crushing the raised dots the brailler creates.

Although Braille notation was designed for folks that square measure blind or visually impaired to scan, before the introduction of the Perkins Brailler, writing Braille was a cumbersome method. Braille writers created Braille characters with a stylus and slate (as developed by prizefighter Braille) or by victimization one among the advanced, expensive, and fragile Braille writing machines on the market at the time.

Fig 2.2

A new version of the Perkins Brailler, the sensible Brailler, was fictitious by David S. Morgan and free in 2011. The sensible Brailler relies on the mechanical action of the classic Perkins Brailler, and, once unpowered, is operable as a customary Brailler. The sensible Brailler includes sensors capturing the mechanical motion of the embosser, and, once supercharged, adds text-to-speech audio feedback and an alphanumeric display to be used by each argus-eyed and blind people. Computer code for the sensible Brailler includes multi-lingual speech and Braille support, as well as English, UK English, Arabic, French, German, Italian, Portuguese, Polish, Russian, and Turkish.

Talkback Braille app

Talkback Braille keyboard could be a new virtual Braille keyboard integrated directly into robot. It’s a quick, convenient thanks to sort on your phone with none extra hardware, whether or not you’re posting on social media, responding to a text, or writing a quick email. As a part of our mission to form the world’s info universally accessible, we have a tendency to hope this keyboard will broadly speaking expand Braille acquisition and exposure among blind and low vision individuals.

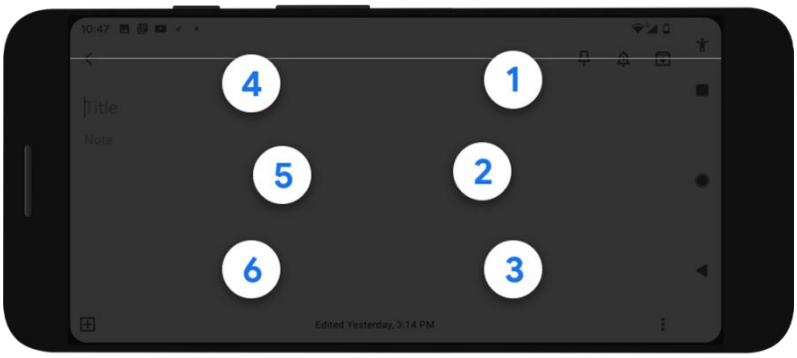

Fig 2.3

The keyboard will be used anyplace you'd commonly sort and permits you to delete letters and words, add lines, and submit text. You'll be able to flip the keyboard on and off as merely as shift between international keyboards. To use the Braille keyboard, activate Talkback within the Accessibility section at intervals Settings, and follow these directions to line it up. Once you got wind of the keyboard, use 3 fingers to swipe au courant your screen and check out active with the gestures tutorial.

2.2 Designing of Device

1. Block Diagram

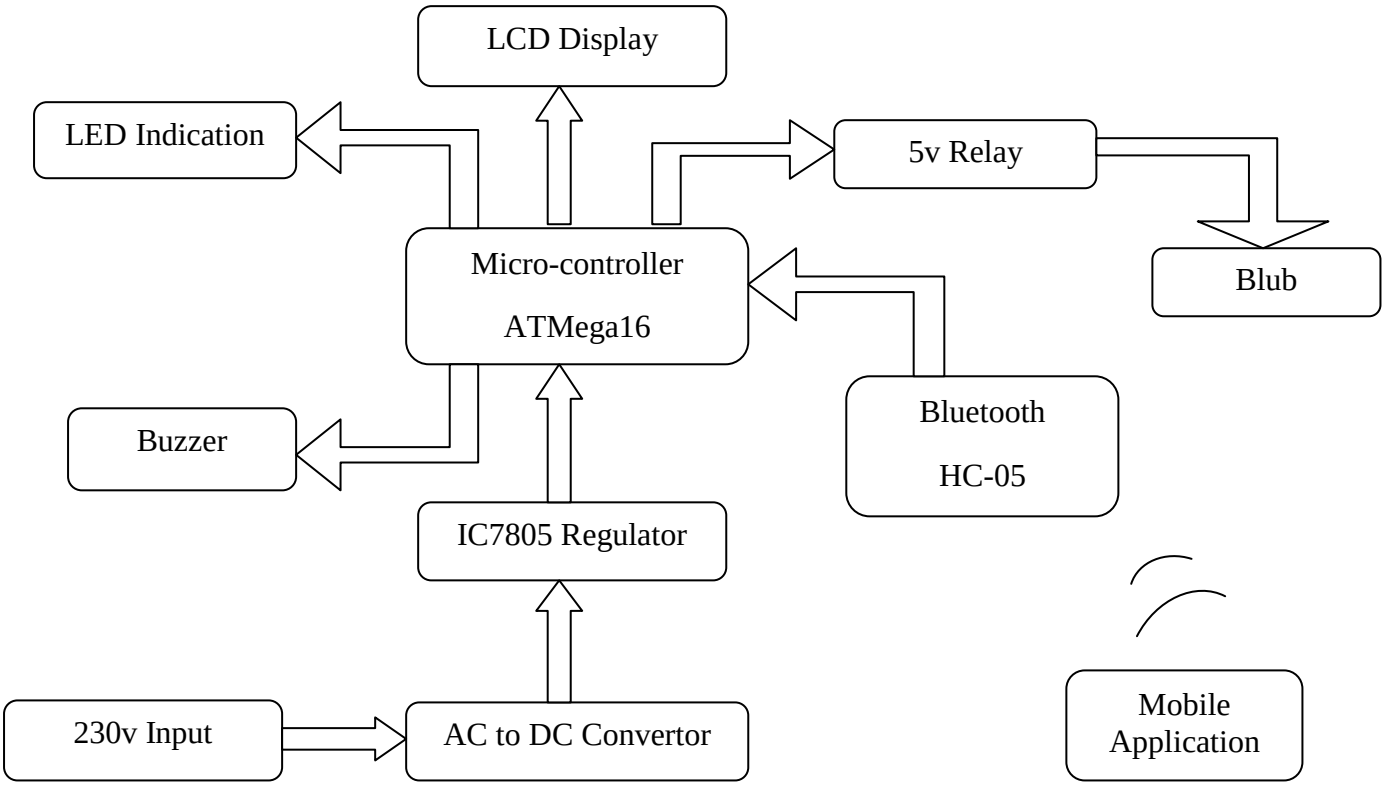

Fig 2.4

Description

1. Firstly, input of 230v as a power will given to the rectifier
2. IC7805 regulator will then provide 5volt to the Microcontroller ATMega16
3. LED will then indicate the proper supply of power by glowing.
4. Buzzer will give indication of sound function.
5. As the circuit gets started,establish the Bluetooth connection between mobile application and Bluetooth module.
6. Then, speech input is given in application and then process by the controller.
7. Controller will convert the speech into Braille Language and display on LCD.
8. Then the power from controller will give to relay and it will glow the Blub.

2.3 Methodology

1. Power Supply

A power provides could be a device that provides electrical power to associate in nursing electrical load. The term is most typically applied to electrical power converters that convert one variety of power to a different, though it's going to additionally seek advice from devices that convert another variety of energy (mechanical, chemical, solar) to power. A regulated power provide is one that controls the output voltage or current to a selected worth; the controlled value is control nearly constant despite variations in either load current or the voltage provided by the facility supply's energy supply.

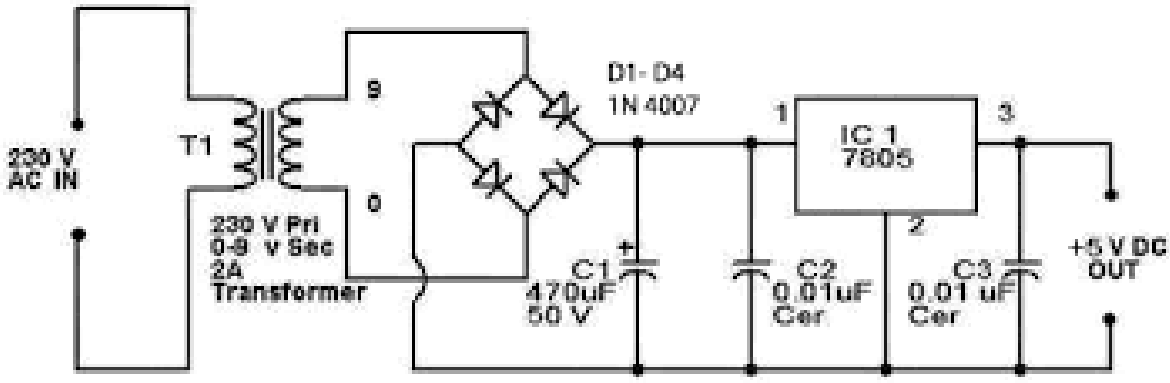

Fig 2.5

TRANSFORMER

Transformer works on the basis of ELECTROMAGNETIC INDUCTION and they are mainly classified into two:

- STEPUP TRANSFORMER
- STEPDOWN TRANSFORMER

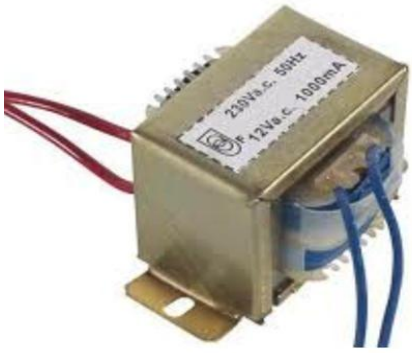

Fig 2.6

Step-up transformer up-convert the input voltage whereas transformer down-converts. For a DC Power supply we've got to use decrease transformers, to convert the high voltage AC provide to low voltage DC.

Transformer plays a very important role in equipment. AC Associate in Nursing DC voltage in Power provide instrumentality square measure virtually achieved by electrical device's transformation and commutation PRODUCT DESCRIPTION A transformer is an device that transfers power between 2 or a lot of circuits through magnetic force induction. Magnetic force induction produces Associate in nursing voltage at intervals a conductor that is exposed to time varied magnetic fields. Transformers square measure won't to increase or decrease the alternating voltages in electrical power applications. It's a step down electrical device during which the secondary coil is quite coil. Thanks to these windings it will ready to step down the voltage. An electrical device changes electricity from high to low voltage or low to high voltage victimization two properties of electricity.

FEATURES

- Output current:1A
- Supply voltage: 220-230VAC
- Output voltage: 12VAC
- Soft Iron Core
- 1Amp Current Drain

RECTIFIER

Rectifiers are wont to convert the curved AC voltage to non-sinusoidal beating DC. The most elements utilized in Rectifiers are diodes because of its shift action. They'll conduct Current solely in one direction, thence the voltage. Therefore we will use them on rectifiers to create the electrical energy one-way.

Rectifiers are classified into Three:-

- HALF WAVE RECTIFIERS
- FULL WAVE RECTIFIERS

- BRIDGE RECTIFIERS

FILTERS

Voltage Regulators are wont to regulate the output Voltage over load. They create the Voltage unvaried with load connected to that. This can eliminates the remaining ripples from the filter output. The output from transformer is also the specified DC. Voltage Regulators includes some safety measures like Current Limiting, tangency etc.

VOLTAGE REGULATOR

Voltage Regulators are wont to regulate the output Voltage over load. They create the Voltage unvaried with load connected to that. This can eliminates the remaining ripples from the filter output. The output from transformer is also the specified DC. Voltage Regulators includes some safety measures like Current Limiting, tangency etc.

- **Brief description of operation:** Gives out will-regulated +5V output, output current capability of 500mA.
- **Circuit protection:** Built-in overheating protection shuts down output when regulator IC gets too hot.
- **Circuit complexity:** Very simple and easy to build.
- **Circuit performance:** Very stable +5V output voltage, reliable operation.
- **Availability of components:** Easy to get, uses only very common basic components.
- **Design testing:** Based on datasheet example circuit, we have used this circuit successfully as part of many electronics projects
- Works on 12v dc input

7805-IC:

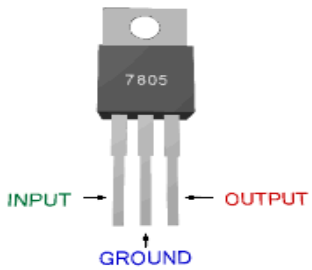

Fig 2.7

- This is the IC which is used for providing the 5V-dc supply to the ATmega16.
- This is connecting to the ATmega16 for getting the 5V.
- This IC has total three pin and it is connect to the 12V dc.
- Power supply to getting the voltage of the 5V dc supply.

2. ATMega16 Microcontroller

DEFINITION:

A microcontroller may be a chip that features a laptop processor with all its support operate (clock & reset), memory (both program and data), and I/O (including bus interface) engineered into the device. These in-built functions minimize the requirement for external circuits and devices to be designed within the final application.

The ATmega16 may be a low-power CMOS 8-bit microcontroller supported the AVR increased reduced instruction set computer architecture} architecture. By death penalty powerful directions in an exceedingly single clock cycle, the ATmega16 achieves throughputs approaching one million instructions per second per megacycle per second permitting the system designed to optimize power consumption versus process speed.

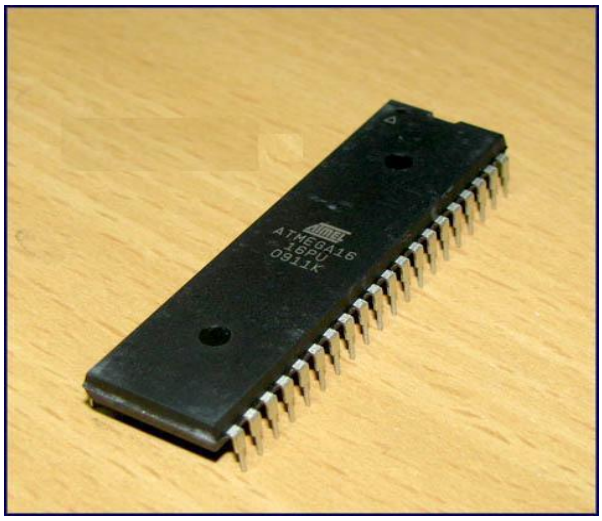

Fig 2.8

The AVR core combines a chic instruction set with thirty two general purpose operating register. All the thirty two registers are directly connected to the Arithmetic Logic Unit (ALU), permitting 2 freelance registers to be accessed in one single instruction dead in one clock cycle. The ensuing design is additional code economical whereas achieving throughputs up to 10 times quicker than standard CISC microcontrollers.

The ATmega16 AVR is supported with a full suite of program and system development tools including: C compilers, macro assemblers, program debugger/simulators, in-circuit emulators, and analysis kits. The boot program will use any interface to transfer the appliance program within the Application nonvolatile storage.

Software within the Boot Flash section can still run whereas the appliance Flash section is updated, providing true Read-While-Write operation. By combining Associate in Nursing 8-bit reduced instruction set computer central processing unit with In-System Self-Programmable Flash on a monolithic chip, the Atmel ATmega16 may be a powerful microcontroller that has a highly-flexible and efficient answer to several embedded management applications.

FEATURES:

• High-performance, Low-power AVR® 8-bit Microcontroller

•Advanced reduced instruction set computer design

•131 Powerful directions – Most Single-clock Cycle Execution

•32 x eight General Purpose operating Registers

•Fully Static Operation

•Up to sixteen million instructions per second output at sixteen megacycles per second

•On-chip 2-cycle number

•Non-volatile Program and information reminiscences

•16K Bytes of In-System Self-Programmable Flash

•Optional Boot Code Section with freelance Lock Bits

In-System Programming by On-chip Boot Program

- True Read-While-Write Operation
- 512 Bytes EEPROM

Endurance: 100,000 Write/Erase Cycles

• 1K computer memory unit Internal SRAM

•Programming Lock for computer code Security

•JTAG (IEEE std. 1149.1 Compliant) Interface

•Boundary-scan Capabilities per the JTAG normal

•Extensive On-chip right Support

• Programming of Flash, EEPROM, Fuses, and Lock Bits through the JTAG Interface

Peripheral options

- Two 8-bit Timer/Counters with Separate Pre scalars and Compare Modes
- One 16-bit Timer/Counter with Separate Pre scalar, Compare Mode, and Capture Mode.
- Real Time Counter with Separate generator
- Four PWM Channels
- 8-channel, 10-bit ADC
- 8 Single-ended Channels
- 7 Differential Channels in TQFP Package solely
- 2 Differential Channels with Programmable Gain at 1x, 10x, or 200x
- Byte-oriented Two-wire Serial Interface
- Programmable Serial USART
- Master/Slave SPI Serial Interface

- Programmable Watchdog Timer with Separate On-chip generator
- On-chip Analog Comparator

Special Microcontroller Features

- Power-on Reset and Programmable Brown-out Detection
- Internal Calibrated RC Oscillator
- External and Internal Interrupt Sources
- Six Sleep Modes: Idle, ADC Noise Reduction, Power-save, Power-down, and Standby andExtended standby.

I/O and Packages

- 32 Programmable I/O Lines
- 40-pin PDIP, 44-lead TQFP, and 44-pad MLF

Operating Voltages

- 2.7 - 5.5V for ATmega16L
- 4.5 - 5.5V for ATmega16

Speed Grades

- 0 - 8 MHz for ATmega16L
- 0 - 16 MHz for ATmega16

PIN DIAGRAM

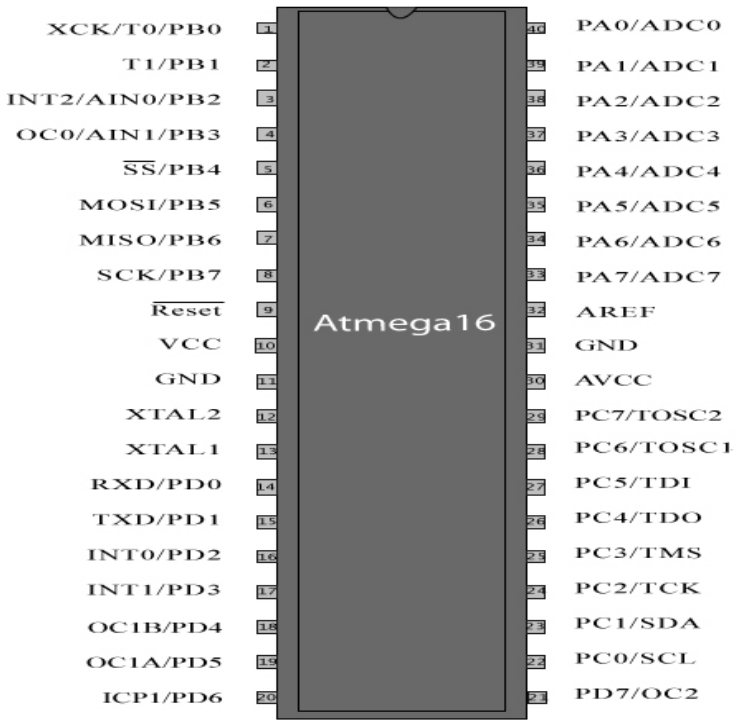

Fig 2.9

PIN DESCRIPTION

- **VCC:** Digital supply voltage.

- **GND**: Ground.

- **Port A (PA7.PA0):** Port A is the analog inputs to the A/D device. Port A additionally is associate 8-bit bi-directional I/O port, if the A/D device isn't used. Port pins will give internal pull-up resistors (selected for every bit). The Port A output buffers have Symmetrical drive characteristics with each high sink and supply capability. Once pins PA0 to PA7 area unit used as inputs and area unit outwardly force low, they'll supply current if the interior pull-up resistors area unit activated.

- **Port B (PB7.PB0):** Port B is associate 8-bit bi-directional I/O port with internal pull-up resistors (selected for every bit). The Port B output buffers have symmetrical drive characteristics with each high sink and supply capability. As inputs, Port B pins that area unit outwardly force low can supply current if the pull-up resistors area unit activated. The Port B pins area unit tri-stated once a reset condition becomes active, albeit the clock isn't running.

- **Port C (PC7.PC0):** Port C is associate 8-bit bi-directional I/O port with internal pull-up resistors (selected for every bit). The Port C output buffers have symmetrical drive characteristics with each high sink and supply capability. As inputs, Port C pins that area unit outwardly force low can supply current if the pull-up resistors area unit activated. The Port C pins area unit tri-stated once a reset condition becomes active, albeit the clock isn't running. If the JTAG interface is enabled, the pull-up resistors on pins PC5 (TDI), PC3 (TMS) and PC2 (TCK) are activated albeit a reset happens.

- **Port D (PD7.PD0):** Port D is associate 8-bit bi-directional I/O port with internal pull-up resistors (selected for every bit). The Port D output buffers have symmetrical drive characteristics with each high sink and supply capability. As inputs, Port D pins that area unit outwardly force low can supply current if the pull-up resistors area unit activated. The Port D pins area unit tri-stated once a reset condition becomes active, albeit the clock isn't running.

- **RESET:** Reset Input. A low level on this pin for longer than the minimum pulse length will generate a reset, even if the clock is not running. Shorter pulses are not guaranteed to generate a reset.

- **XTAL1:** Input to the inverting Oscillator amplifier and input to the internal clock operating circuit.

- **XTAL2:** Output from the inverting Oscillator amplifier.

- **AVCC:** AVCC is the supply voltage pin for Port A and the A/D Converter. It should be externally connected to VCC, even if the ADC is not used. If the ADC is used, it should be connected to VCC through a low-pass filter.

- **AREF:** AREF is the analog reference pin for the A/D Converter.

SOFTWARE DESCRIPTION:

Software style includes series of multidimensional language that helps to trace the entire program and alter straightforward programming and error checking. Software system tools area unit as vital as hardware tools needed to mould an ideal style.

The following area unit the various tools used for software system development.

- Bascom-AVR
- Robokits AVR USB software engineer (STK 500)

BASCOM SOFTWARE:-

Bascom software system is employed to produce you with Bascom development tools for AVR based mostly microcontrollers. With the Bascom tools, you'll generate embedded applications for nearly each AVR spinoff. The supported microcontroller's area unit listed within the Bascom-AVR.

Fig 2.10

All AVR microcontrollers area unit supported Atmel's low-power nonvolatile CMOS technology. The on-chip in-system programmable (ISP), downloadable non-volatile storage permits devices on the user's board to be reprogrammed via SPI or with the assistance of a standard programming device.

By combining the economical design with the downloadable non-volatile storage on an equivalent chip, the AVR microcontrollers represent associate economical approach to applications within the "Embedded Controller" market.

Advantages:

- Easy to interface with totally different devices.
- Easy to know.
- Reduces program length.
- Reduces the quality.
- It is versatile, because it may be worked with alternative microcontroller by creating minimum changes.

ROBOKITS AVR Programmer (STK 500):

We area unit victimization STK-500 loader to load the hex file of the program within the microcontroller ATmega16. It's done by means that of ROBOKITS AVR software engineer. This could even be used for numerous alternative microcontrollers.

Here the snap shot of this software is shown in figure.

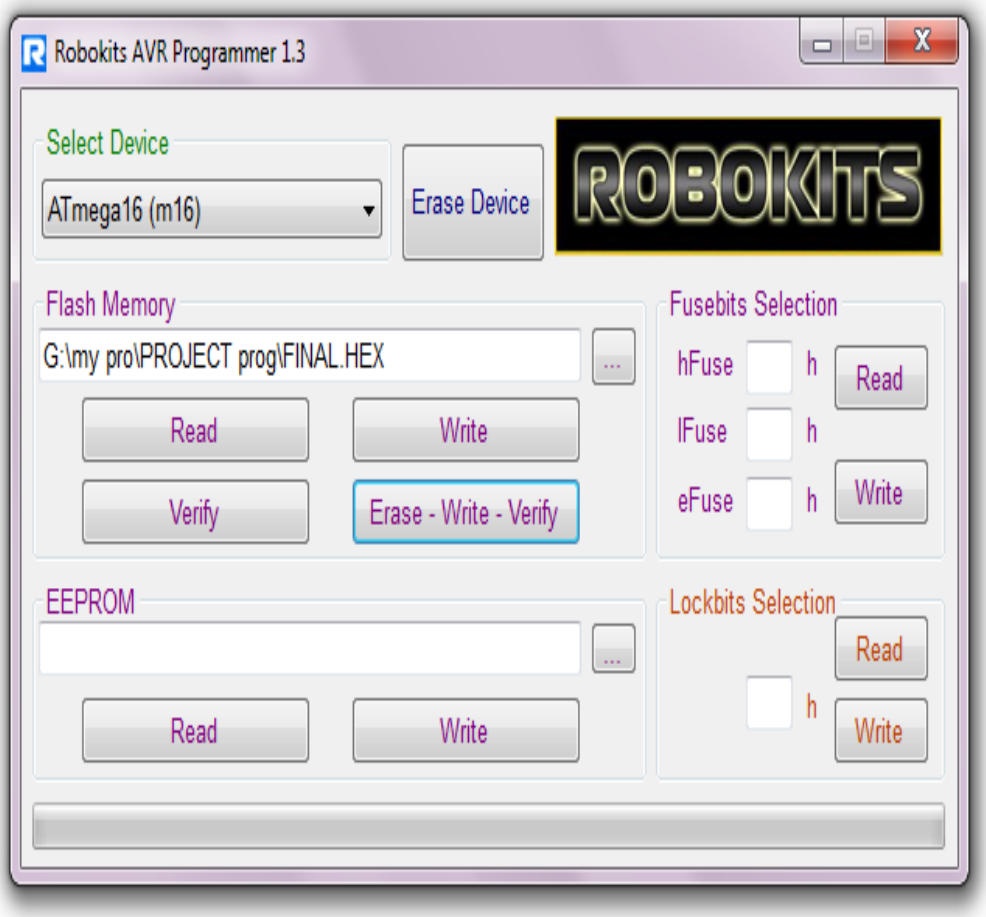

Fig 2.11

3. RELAY INTERFACING

What is a relay?

A relay is associate magnetic force switch that's wont to activate and switch off a circuit by a coffee power signal, or wherever many circuits should be controlled by one signal.

We know that almost all of the high finish industrial application devices have relays for his or her effective operating. Relays area unit easy switches that area unit operated each

electrically and automatically. Relays incorporate associate magnet and additionally a group of contacts. The shift mechanism is allotted with the assistance of the magnet. There are alternative operational principles for its operating. However they dissent per their applications. Most of the devices have the applying of relays.

Why is a relay used?

The main operation of a relay comes in places wherever solely a low-power signal may be wont to management a circuit. It's additionally employed in places wherever only 1 signal may be wont to management heaps of circuits. The applying of relays started throughout the invention of telephones. They compete a vital role in shift calls in phone exchanges. They were additionally employed in long distance telegraphy. They were wont to switch the signal coming back from one supply to a different destination. When the invention of computers they were additionally wont to perform mathematician and alternative logical operations. The high finish applications of relays need high power to be driven by electrical motors so on. Such relays area unit known as contactors.

How relay works?

The relay function can be better understood by explaining the following diagram given below.

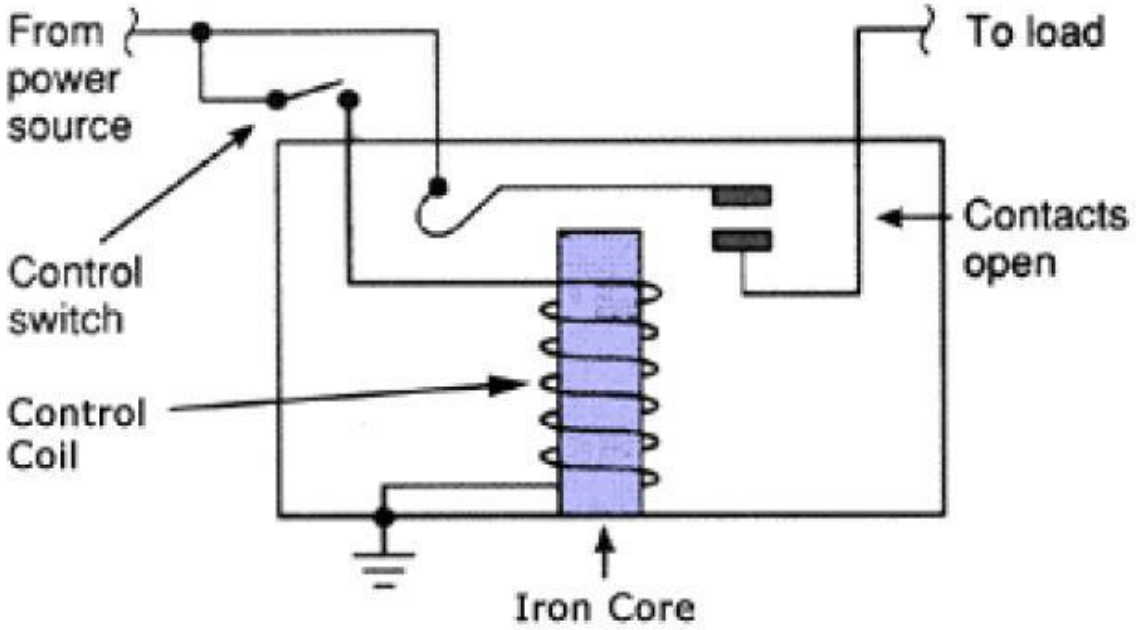

Fig 2.12

Relay Design

The diagram shows associate inner section diagram of a relay. Associate iron core is enclosed by an effect coil. As shown, the facility supply is given to the magnet through an effect switch and thru contacts to the load. Once current starts flowing through the management coil, the magnet starts energizing and so intensifies the force field. So the higher tangency starts to be interested in the lower mounted arm and so closes the contacts inflicting a brief circuit for the facility to the load. On the opposite hand, if the relay was already de-energized once the contacts were closed, then the contact move oppositely associated create a circuit.

As soon as the coil current is off, the movable armature will be returned by a force back to its initial position. This force will be almost equal to half the strength of the magnetic force. This force is mainly provided by two factors. They are the spring and also gravity.

Relays are mainly made for two basic operations. One is low voltage application and the other is high voltage. For low voltage applications, more preference will be given to reduce the noise of the whole circuit. For high voltage applications, they are mainly designed to reduce a phenomenon called arcing.

Relay Basics

The basics for all the relays are a similar. Take a glance at a four – pin relay shown below. There are 2 colors shown. The inexperienced color represents the feedback circuit and also the red color represents the load circuit. Atiny low management coil is connected onto the feedback circuit. A switch is connected to the load. This switch is controlled by the coil within the feedback circuit. Currently allow us to take the various steps that occur in a very relay.

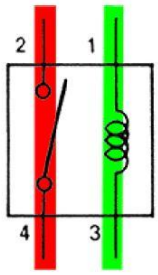

Fig 2.13

Relay operation

Energized Relay (ON)

As shown in the circuit, the current flowing through the coils represented by pins 1 and 3 causes a magnetic field to be aroused. This magnetic field causes the closing of the pins 2 and 4. Thus the switch plays an important role in the relay working. As it is a part of the load circuit, it is used to control an electrical circuit that is connected to it. Thus, when the electrical relay in energized the current flow will be through the pins 2 and 4.

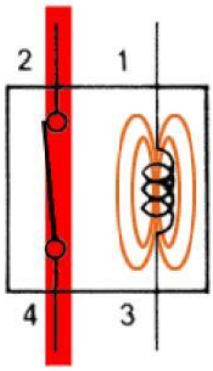

Fig 2.14

Energized Relay (ON)

De – Energized Relay (OFF)

As soon as the current flow stops through pins 1 and 3, the relay switch opens and thus the open circuit prevents the current flow through pins 2 and 4. Thus the relay becomes de-energized and thus in off position.

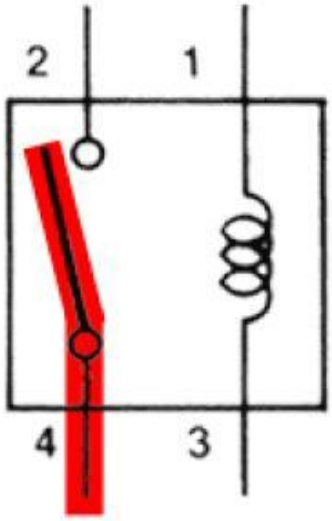

Fig 2.15

De-Energized Relay (OFF)

In simple, when a voltage is applied to pin 1, the electromagnet activates, causing a magnetic field to be developed, which goes on to close the pins 2 and 4 causing a closed circuit. When there is no voltage on pin 1, there will be no electromagnetic force and thus no magnetic field. Thus the switches remain open.

4. LCD Interfacing

16x2 LCD Module

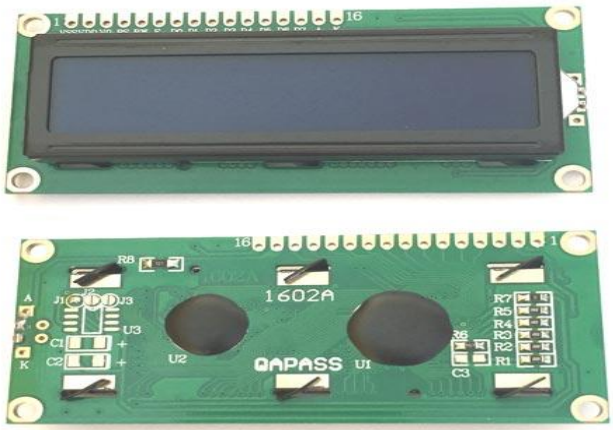

Fig 2.16

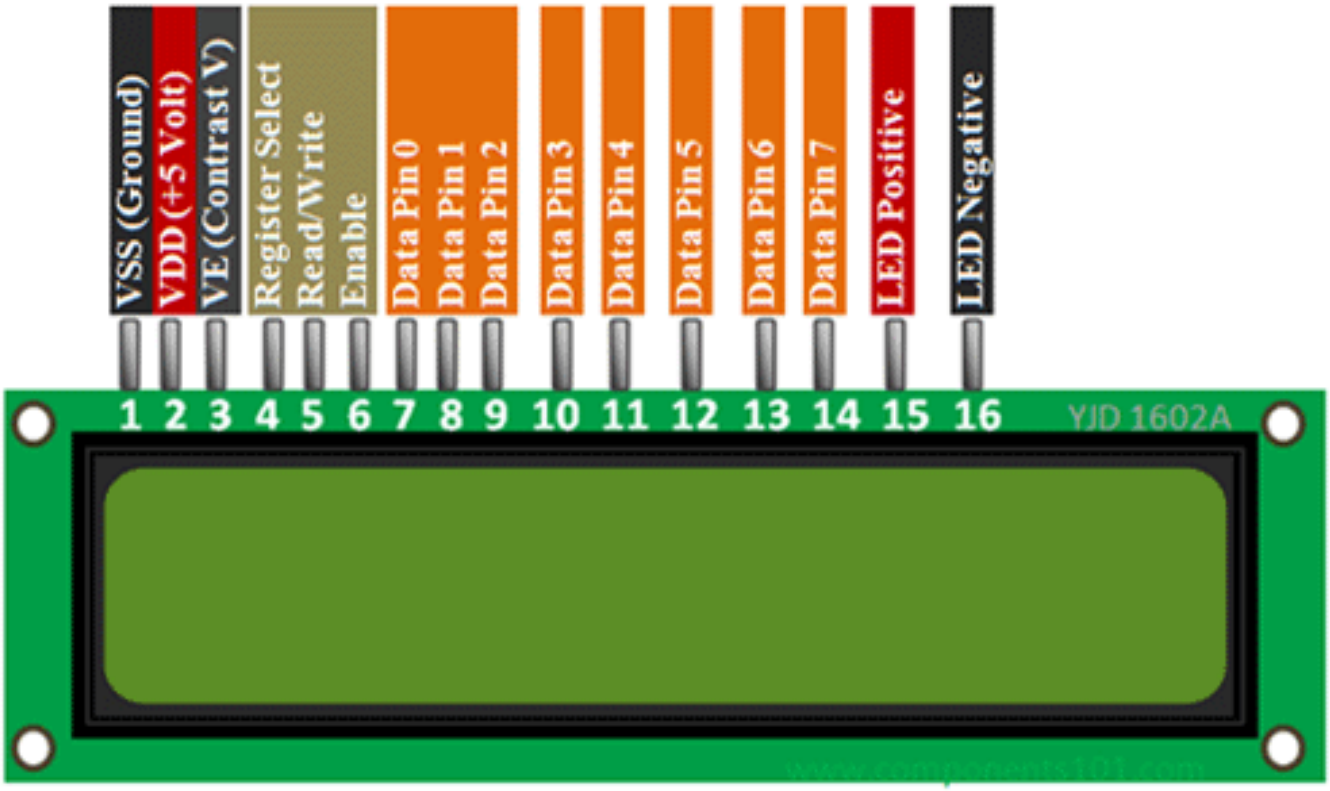

Fig 2.17

Features of 16×2 LCD module

- Operating Voltage is 4.7V to 5.3V
- Current consumption is 1mA without backlight
- Alphanumeric LCD display module, meaning can display alphabets and numbers
- Consists of two rows and each row can print 16 characters.
- Each character is build by a 5×8 pixel box
- Can work on both 8-bit and 4-bit mode
- It can also display any custom generated characters
- Available in Green and Blue Backlight

16×2 LCD is named so because; it has 16 Columns and 2 Rows. There are a lot of combinations available like, 8×1, 8×2, 10×2, 16×1, etc. but the most used one is the 16×2 LCD. So, it will have (16×2=32) 32 characters in total and each character will be made of 5×8 Pixel Dots.

5. Bluetooth HC-05 module

Introduction

- It is employed for several applications like wireless receiver, game controllers, wireless mouse, wireless keyboard and plenty of a lot of client applications.
- It has vary up to <100m that depends upon transmitter and receiver, atmosphere, geographic & urban conditions.

- It is IEEE 802.15.1 standardized protocol, through that one will build wireless Personal space Network (PAN). It uses frequency-hopping unfold spectrum (FHSS) radio technology to send knowledge over air.
- It uses serial communication to speak with devices. It communicates with microcontroller victimization port (USART).

HC-05 Bluetooth Module

- HC-05 could be a Bluetooth module that is intended for wireless communication. This module is utilized in a master or slave configuration.

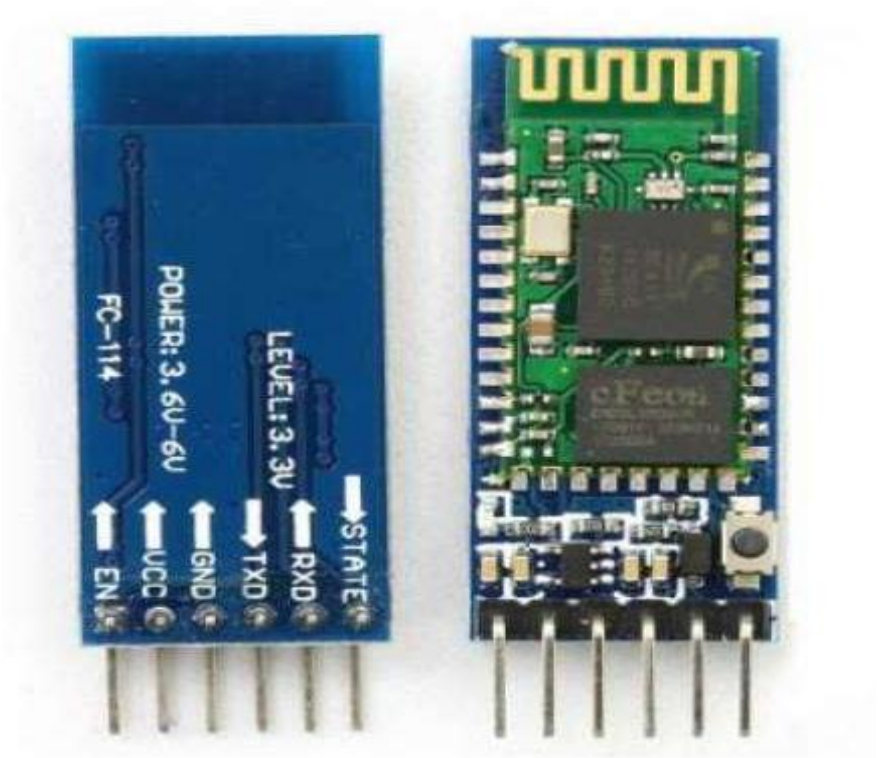

Fig 2.18

HC-05 Bluetooth Module

Pin Description

Fig 2.19

Bluetooth serial modules permit all serial enabled devices to speak with one another victimization Bluetooth.

It has half-dozen pins,

1. Key/EN: it's wont to bring Bluetooth module in AT commands mode. If Key/EN pin is about to high, then this module can add command mode. Otherwise by default it's in knowledge mode. The default information measure of HC-05 in command mode is 38400bps and 9600 in knowledge mode.
   HC-05 module has 2 modes,
a. Knowledge mode: Exchange of knowledge between devices.
b. Command mode: It uses AT commands that are wont to amendment setting of HC-05. To send these commands to module serial (USART) port is employed.
2. VCC: Connect five V or three.3 V to the current Pin.
3. GND: Ground Pin of module.
4. TXD: Transmit Serial knowledge (wirelessly received knowledge by Bluetooth module transmitted out serially on TXD pin)
5. RXD: Receive knowledge serially (received knowledge is transmitted wirelessly by Bluetooth module).
6. State: It tells whether or not module is connected or not.

HC-05 module Information

HC-05 module info

- HC-05 has red semiconductor diode that indicates affiliation standing, whether or not the Bluetooth is connected or not. Before connecting to HC-05 module this red semiconductor diode blinks unceasingly in an exceedingly periodic manner. Once it gets connected to the other Bluetooth device, its blinking slows right down to 2 seconds.
- This module works on three.3 V. we are able to connect five offer voltage further since the module has on board 5 to three.3 V regulator.
- As HC-05 Bluetooth module has three.3 V level for RX/TX and microcontroller will observe three.3 V level, so, no got to shift transmit level of HC-05 module. However we want to shift the transmit voltage level from microcontroller to RX of HC-05 module.

6. Mobile app

Bluetooth voice: Arduino Voice Controller

Fig 2.20

Features:

• You can add any number of custom commands.
• You can enable "Raw Input" to get what you say in serial.
• You can update any command you previously added.
• You can control leds, relay or start your own home automation project.
• You can set the app to auto connect to your devices (like HC 05 or HC 06) for more convenience

Fig 2.21

2.4 Implementation Strategy

- First, problem the society is facing has been concentrated here. We have focused on blind people society.
- We have decided to do something on writing part of the blind. We have tried to make simple for them. So this topic has been selected
- Block diagram has been made by keeping in mind the merits and demerits of it.
- We have then gone for some practical learning of the electronic components such as solderingof components on PCB.
- Implementation of power supply of project is done.
- Interfacing of LED and Buzzer
- Interfacing of ATMega 16 with LCD

3. Implementation

3.1 Actual Implementation

1. Power Supply

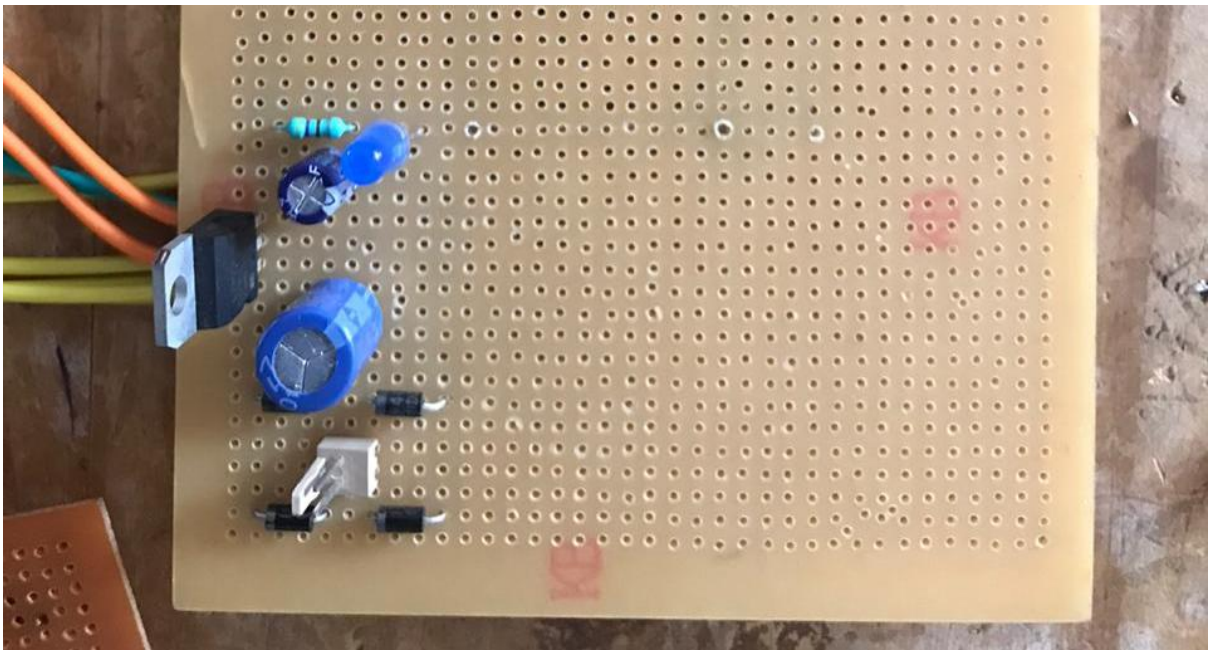

Fig 3.1

This circuit will provide exact 5v dc to microcontroller and LCD.

2. Controller Board

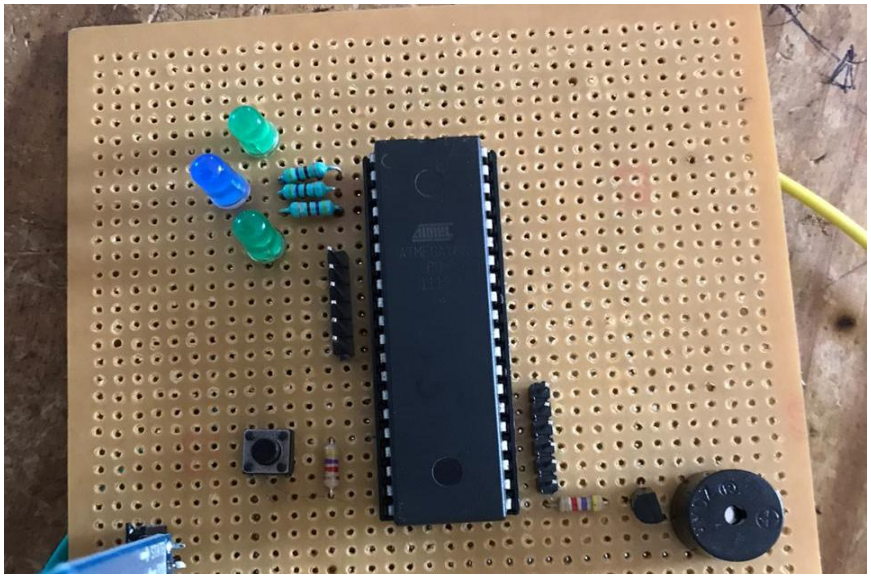

Fig 3.2

This circuit have controller for processing, Buzzer for sound indication and led for start and stop indication.

3. Relay interfacing

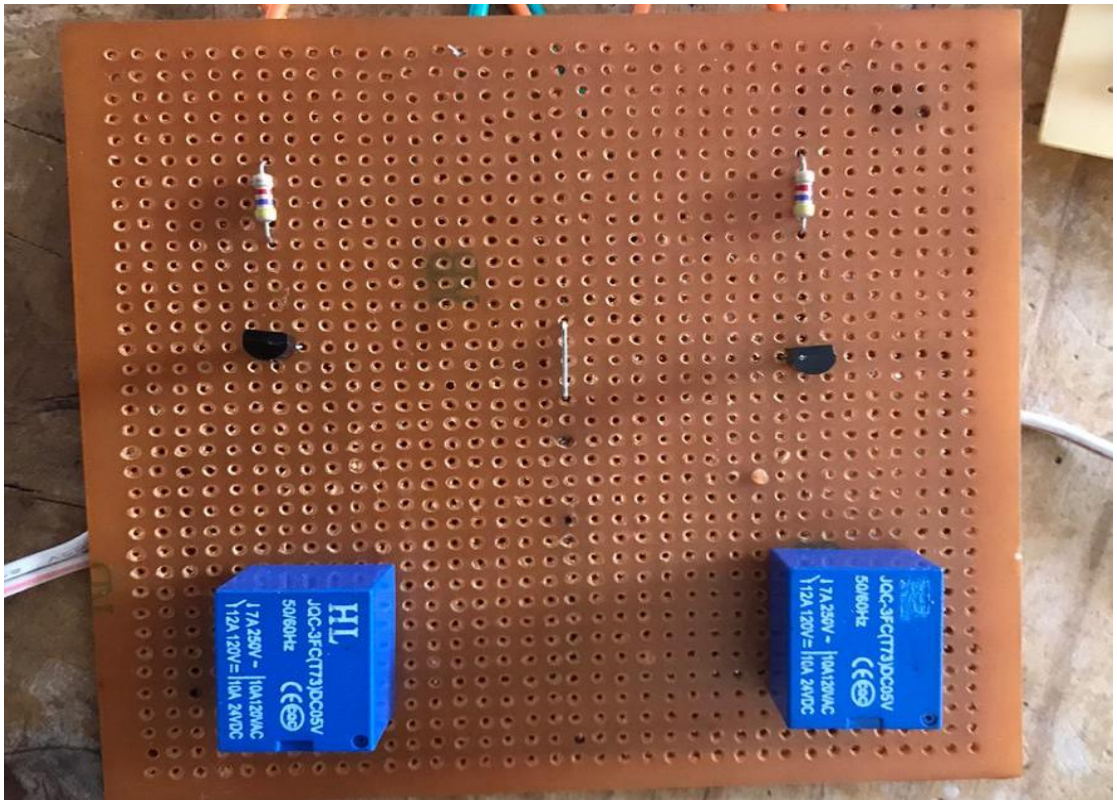

Fig 3.3

This circuit for glowing blub

4. LCD interfacing

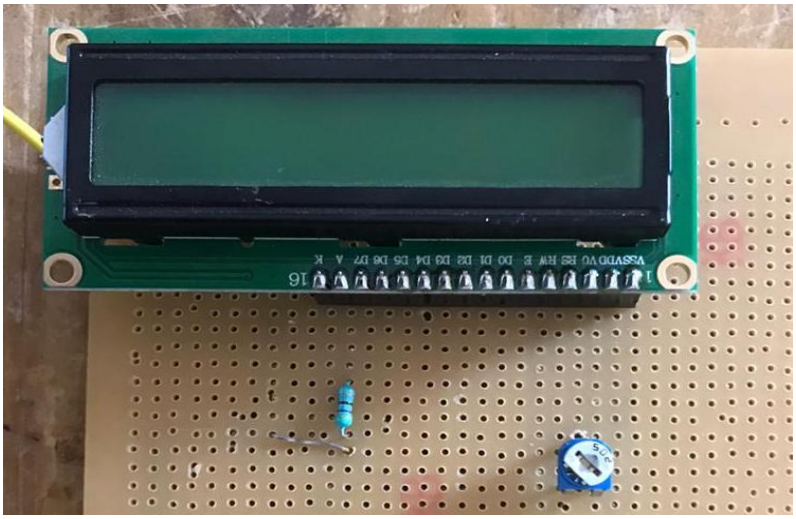

Fig 3.4

This circuit shows the output of the device i.e., Braille code.

5. Bluetooth Interfacing

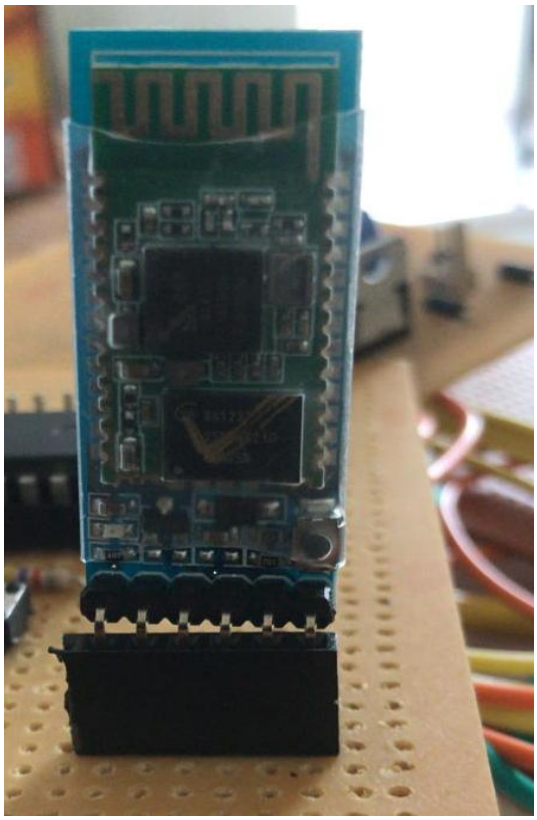

Fig 3.5

This module is for connection to the phone application.

3.2 Project hardware

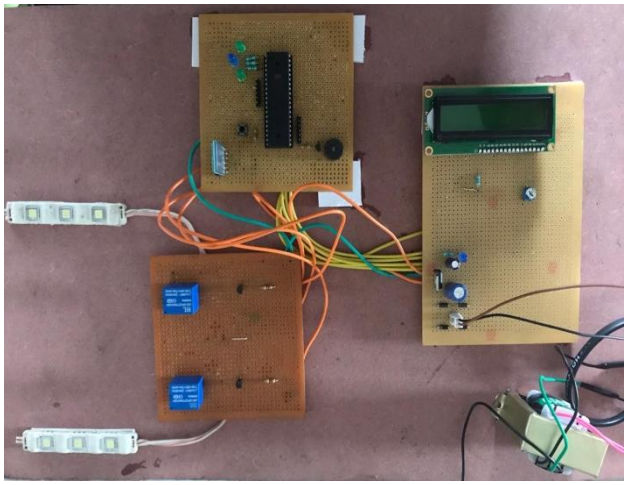

Fig 3.6

3.3 Result

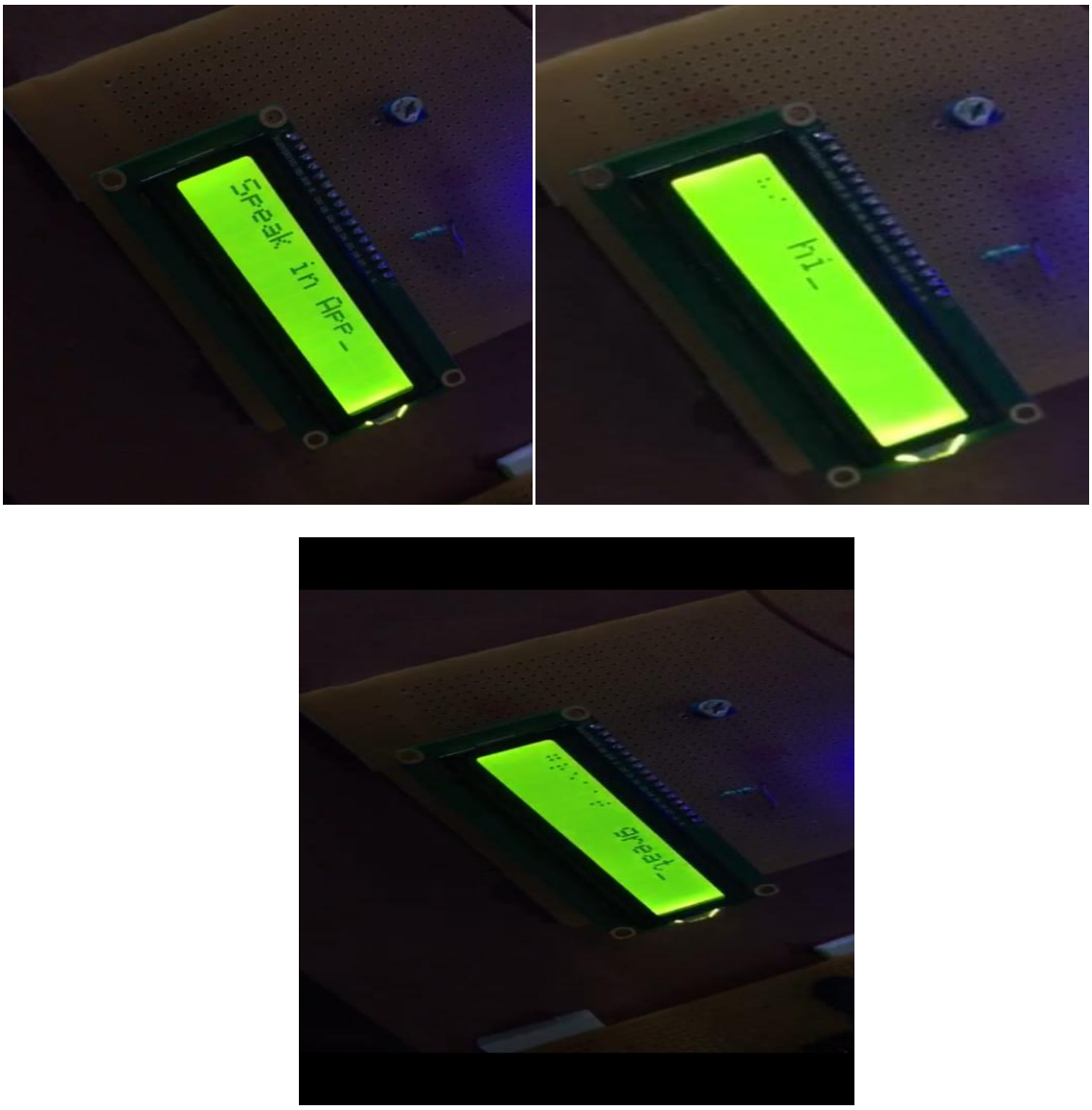

Fig 3.7

This are some results on listening to the voice input from mobile app.

4. Summary

4.1 Advantages

- This device just required voice for input
- It’s very Portable
- Cost efficient
- Blind people will easily write anything without anyone’s help

4.2 Disadvantages

- Speech recognized in small distance
- Noise can fail exact output

4.3 Conclusion and Future Scope

Our research is focused mainly on the developing a system that reduces the difficulty in writing a document and introducing a cognitive system which can give the visually impaired people a sense of haptic feeling. This system cost effective and the satisfaction of the user is tremendous. However, this can be the ultimate solution for the blind people for an effective interaction with writing a document, book or article.

The importance of Braille to the blind community cannot beundermined. It has significantly contributed to individuals,who have this type of disability, functioning normally world and has provided them with the opportunity to maketheir society better, through their very valuable contributionsin many different areas of everyday life.Now those people can learn themselves using this standalonekit. It is easy one to use.

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